

VISIORA

AI Based Property Image Classification
Smart Visual Organization for Real Estate Platforms

TEAM HRY2

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Datathon Hackathon
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Abstract

Real estate platforms receive large numbers of property images uploaded by different sellers. These images are often unorganized and lack proper categorization, making it difficult for potential buyers to browse and search effectively.

VISIORA is an Artificial Intelligence based system designed to automatically analyze and organize property images. The system detects room types, identifies furniture objects, generates descriptive tags, and categorizes images intelligently.

The platform supports batch uploads, folder uploads, session based category browsing, and visual analytics dashboards that help users manage property images efficiently.

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1 Introduction

The growth of real estate platforms has resulted in an increasing volume of property images being uploaded daily. These images often lack structured metadata such as room type or furniture information. As a result, users searching for specific property features experience difficulty locating relevant images.

VISIORA addresses this problem by providing an AI powered platform that automatically classifies property images and organizes them into meaningful categories.

1.1 Problem Statement

Real estate platforms face several challenges in managing property images:

- Images are uploaded without proper categorization.
- Manual tagging is inefficient and time consuming.
- Large image collections are difficult to navigate.
- Buyers struggle to find specific room images.

1.2 Objectives

The main objectives of VISIORA include:

- Automatically classify property images by room type.

- Detect furniture objects present in the image.
- Generate intelligent tags for better organization.
- Support batch and folder image uploads.
- Provide visual analytics for analyzed data.

2 System Architecture

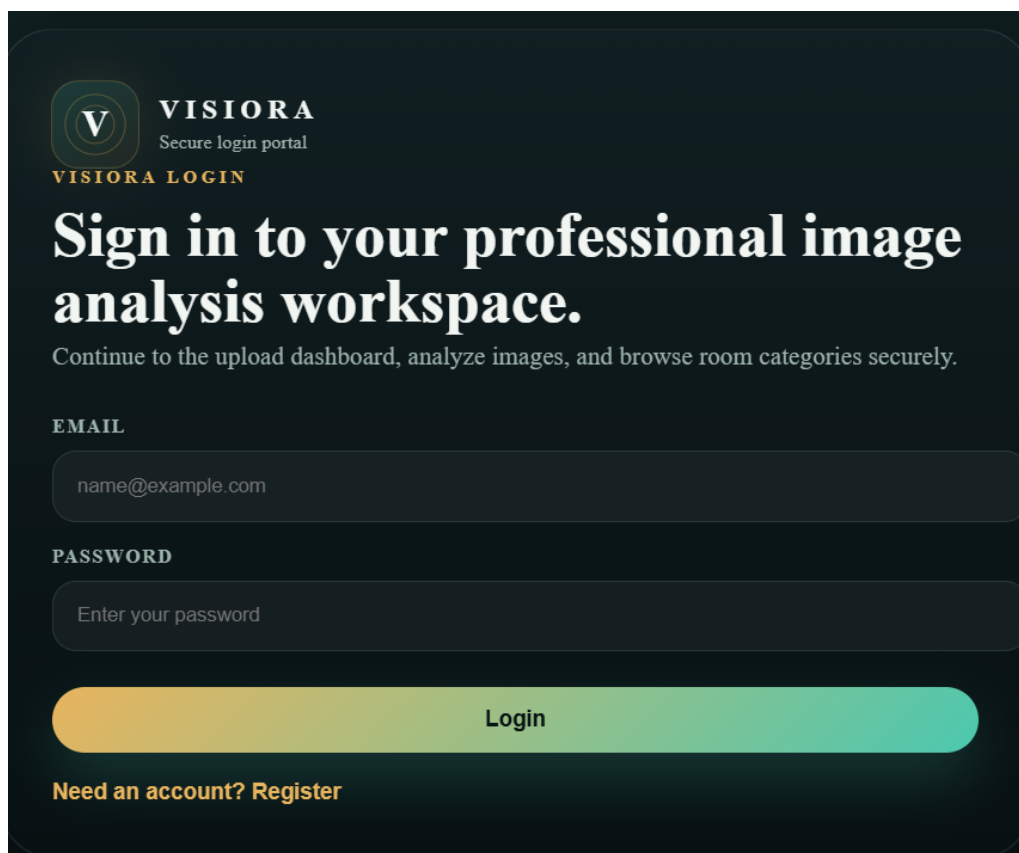
VISIORA follows a modular architecture consisting of multiple components:

- React based frontend interface
- Node.js backend API
- FastAPI AI service
- MongoDB database

The frontend allows users to upload and view images. The backend communicates with the AI service to analyze images and store results in MongoDB.

3 System Interface

3.1 Login Interface



The image shows a dark-themed login interface for VISIORA. At the top left is the VISIORA logo, a stylized 'V' inside a circle, followed by the text 'VISIORA' and 'Secure login portal' below it. Below this is the text 'VISIORA LOGIN' in a smaller, orange font. The main heading is 'Sign in to your professional image analysis workspace.' in a large, white, serif font. Below the heading is a subtitle in a smaller, white, sans-serif font: 'Continue to the upload dashboard, analyze images, and browse room categories securely.' The login form consists of two input fields: 'EMAIL' with a placeholder 'name@example.com' and 'PASSWORD' with a placeholder 'Enter your password'. Both fields have a light gray border. Below the password field is a large, rounded rectangular button with a gradient from orange to green, labeled 'Login' in white. At the bottom left of the form is a link that says 'Need an account? Register' in orange text.

Figure 3.1: VISIORA secure login page

The login interface ensures secure access to the VISIORA platform.

3.2 Main Dashboard

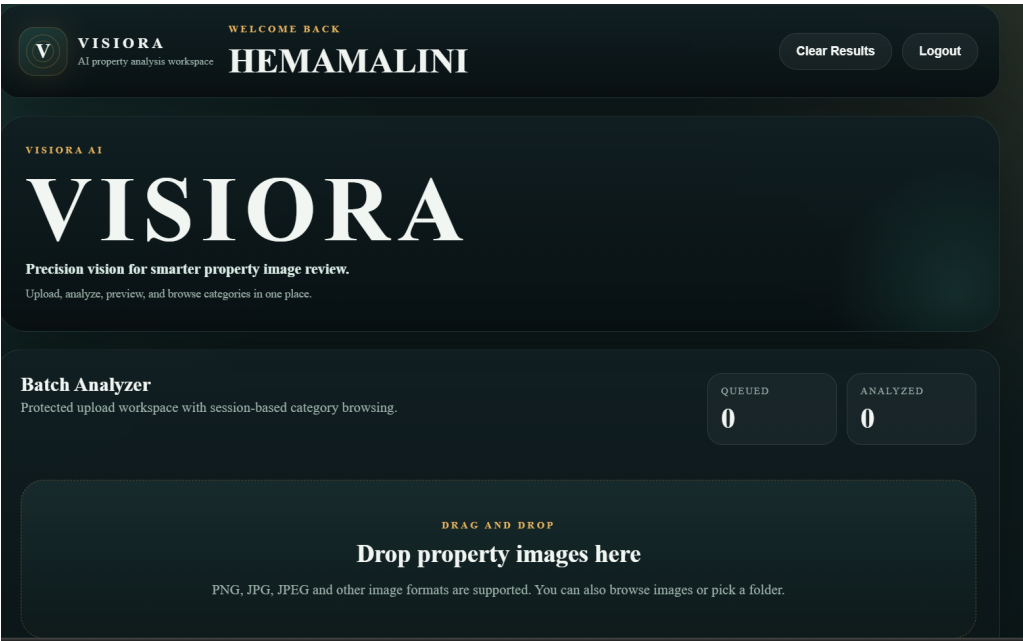


Figure 3.2: VISIORA main dashboard

The dashboard provides access to image analysis tools and analytics.

3.3 Batch Image Upload

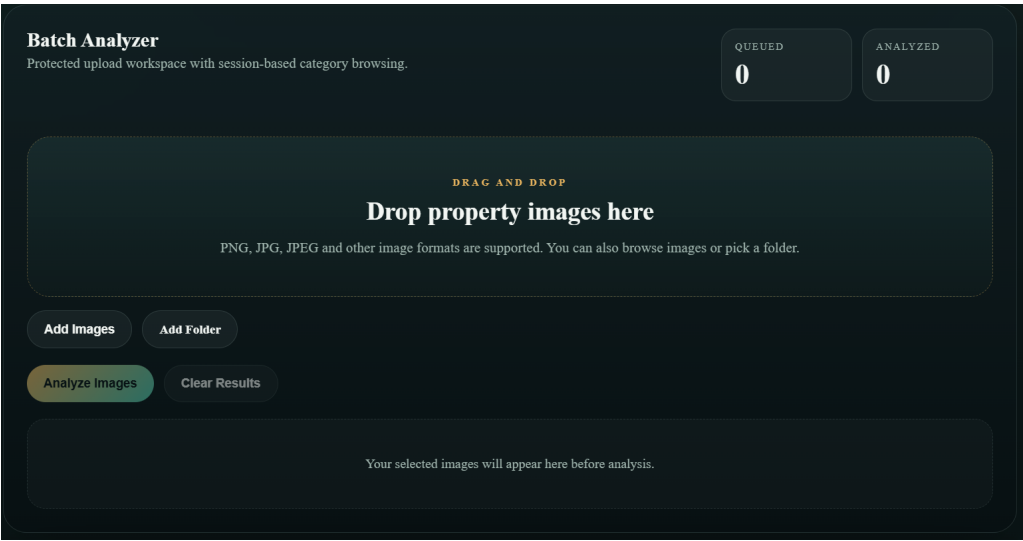


Figure 3.3: Batch image upload interface

Users can upload multiple images or folders using drag and drop functionality.

3.4 Analytics Dashboard

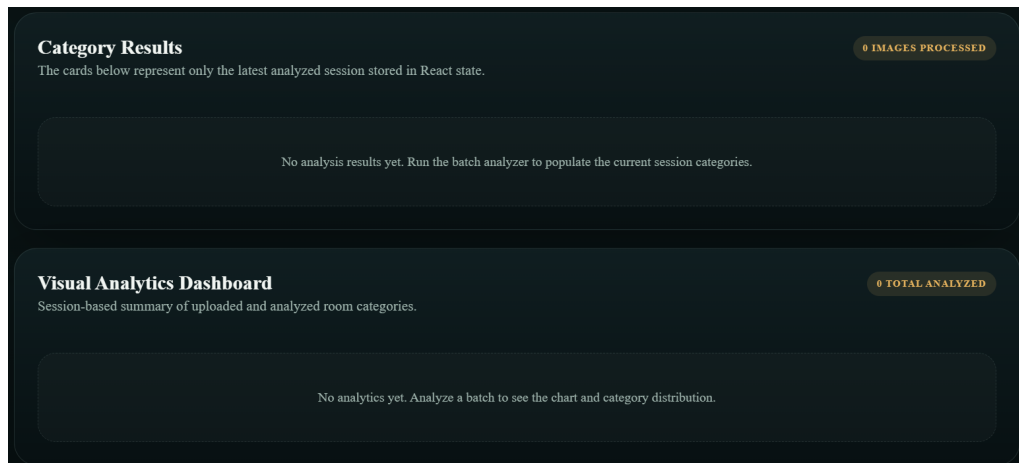


Figure 3.4: VISIORA analytics dashboard

The analytics dashboard shows insights about analyzed images and their distribution.

4 Technology Stack

VISIORA is developed using the following technologies:

- React.js – Frontend user interface
- Node.js and Express.js – Backend API
- FastAPI – AI inference service
- PyTorch – Deep learning framework
- MongoDB – Database for storing results
- JWT authentication – Secure login system

5 Conclusion

VISIORA demonstrates how artificial intelligence can improve property image organization in real estate platforms. By automatically classifying images, detecting furniture objects, and organizing results into categories, the system enhances user experience and improves efficiency.

The proposed platform can be extended further with additional AI features such as caption generation, duplicate detection, and automated property description generation.